

Step-by-step causal analysis of antibiotic continuation in ICU sepsis: emulating a target trial from MIMIC-IV

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Abstract

Antibiotic over-prescription in the ICU drives microbial resistance, *C. difficile* colitis, and avoidable adverse drug effects, yet evidence for when to stop, narrow, or continue broad-spectrum therapy at the 72 h mark is scarce. We instantiate a five-step causal-inference framework (study design, identification, estimation, vibration analysis, validation & guardrails) to extract decision-support evidence on this question from MIMIC-IV. We emulate a three-arm target trial (*continue* / *de-escalate* / *cease*) at 72 h after the first broad-spectrum order in 9,331 sepsis-3 patients. The estimation step employs double machine learning (DML) and doubly-robust learners with logistic mortality models and random-forest propensity models; the vibration analysis sweeps estimator $\times$ feature-set $\times$ treatment-window $\times$ permutation-label choices. Validity is probed with five orthogonal diagnostics: positivity (propensity overlap), per-arm calibration, E-value, K-fold CI coverage, and a Mahalanobis out-of-distribution check. Across 58 sensitivity fits, the *cease vs. continue* contrast is the only estimate that survives every probe: ATE ≈ -3 to -4 percentage points on 28-day mortality (15/18 fits with CI strictly below zero, E-value 1.55, permutation null $|\overline{ATE}| < 1.7$ pp). The *de-escalate* contrasts are null across all diagnostics; positivity (overlap 36 %) and per-arm calibration (slope 0.34) flag them as untrustworthy at the individual level. We deploy the model in a clinical-decision-support frontend that surfaces overlap and out-of-distribution guardrails alongside every estimate. The paper is intended as a worked replication template for stewardship-style EHR causal-inference questions.

Author summary

A clinician at the 72-hour point of a sepsis admission has three options for the broad-spectrum antibiotic that was started empirically: continue it, narrow it to a culture-guided agent, or stop antibiotics. The decision is genuinely causal — it concerns what *would happen* under each option for this specific patient — yet most ICU prediction models conflate prognosis with effect. We re-frame the question as a three-arm target trial in MIMIC-IV, equip the estimator with a battery of identification-validity diagnostics borrowed from pharmaco-epidemiology (E-value, positivity, calibration, permutation null), and report what survives. The headline finding is that the only contrast with replicable causal signal is *ceasing* versus continuing broad-spectrum therapy; the de-escalate arm is too sparse and too poorly calibrated in MIMIC-IV to support individual-level recommendations. We deliberately keep the implementation accessible: every step is wired into a reproducible pipeline, and every confidence interval is paired with a diagnostic that says when not to trust it.

1 Introduction — antibiotic-continuation decisions need causal answers

Surviving Sepsis Campaign guidelines recommend that empirical broad-spectrum antibiotics begun on ICU arrival be re-evaluated at 48–72 h, and de-escalated or stopped where possible [1]. The recommendation is broadly accepted; what it means *for an individual patient* is not. The 72-hour decision is a textbook “confounding by indication” problem: the patient’s measured trajectory — fever curve, lactate clearance, blood-culture growth, SOFA delta — informs both the antibiotic decision and the outcome. Naïve comparisons of mortality by 72-hour treatment arm therefore overstate the benefit of continuing therapy (sicker-looking patients are continued, and sicker patients die) and conflate the effect of antibiotics with the effect of severity.

Causal inference frameworks are designed for exactly this scenario. Doutréline et al. [2] laid out a five-step protocol — study design, identification, estimation, vibration analysis, and treatment-heterogeneity analysis — for extracting average treatment effects (ATEs) from EHR data while making each analytic choice auditable. They illustrated the framework on the binary question of *albumin vs. crystalloids* in septic shock. In this work we apply the same scaffolding to a three-arm, time-anchored stewardship question: *continue broad-spectrum, de-escalate to narrow-spectrum, or cease antibiotics at 72 hours?* We extend the original framework with four explicit validity probes (overlap, calibration, E-value, permutation null) and one user-interface guardrail (Mahalanobis OOD), and bundle everything into a reproducible pipeline.

2 Framework

Figure 2. Step-by-step framework applied to the antibiotic-continuation question

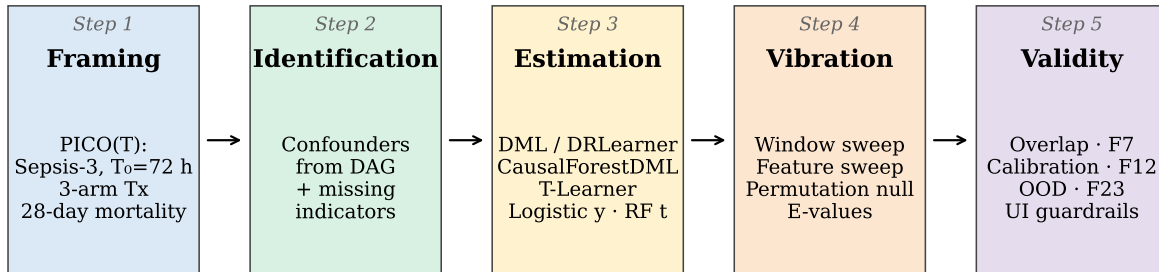


Figure 1: The five-step analytic pipeline used for the antibiotic-continuation question. Each step has its own audit artefacts and diagnostics; the design refuses to display an ATE for which the matching diagnostic is missing.

2.1 Step 1 — Study design & PICO(T) framing

The medical decision is recast as a target trial. The PICO(T) components are:

- **Population:** adult ICU admissions with a sepsis-3 flag and an ICU length of stay ≥ 72 h.
- **Intervention** (three-arm): at the 72-hour decision point, (0) continue broad-spectrum, (1) de-escalate to a narrow-spectrum agent only, or (2) cease all antibiotics.
- **Comparator:** pairwise contrasts between any two arms.

- **Outcomes:** 28-day mortality (primary); ventilator-free days at 28 (VFD-28), vasopressor-free days, ICU length of stay, AKI worsening, secondary blood-culture infection, and *C. difficile* therapy initiation as proxy for CDI (secondary).
- **Time:** $T_0 = 72$ h after the first broad-spectrum antibiotic order. The classification of treatment arm uses a ± 12 h window around T_0 .

Immortal-time bias. Patients who die between hours 0 and 72 cannot be in any arm; restricting the cohort to those alive in the ICU at T_0 removes the bias. **Selection bias.** A single patient can have multiple eligible ICU stays; we retain only the earliest qualifying stay per `subject_id` (within-subject correlation otherwise inflates effective n and biases CIs narrow).

Figure 1. Cohort selection flow

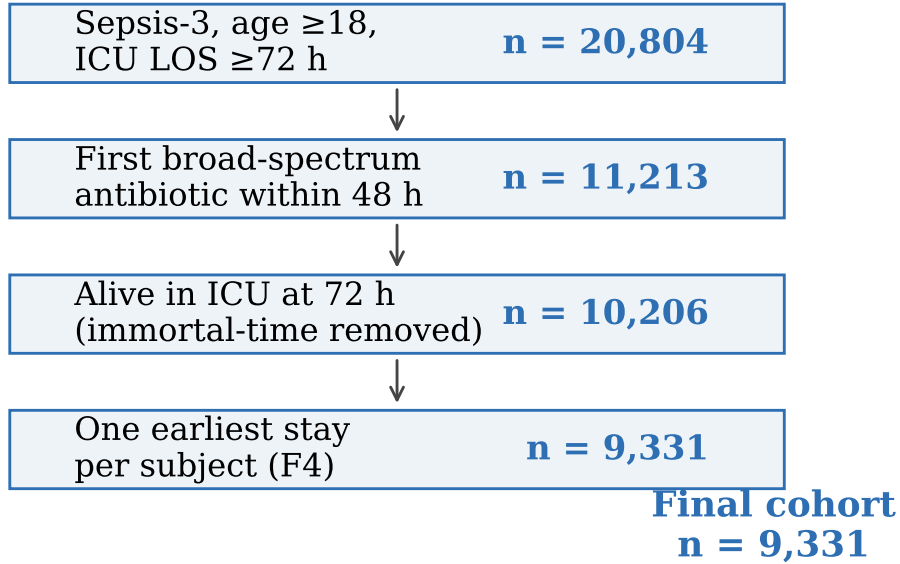


Figure 2: Cohort flow. From 20,804 sepsis-3 stays in MIMIC-IV, 9,331 enter the analytic cohort after applying the broad-spectrum-antibiotic, immortal-time, and one-stay-per-subject filters.

2.2 Step 2 — Identification

The identification step lists the variables a causal estimator must condition on to recover an ATE from the joint distribution. We use the DAG encoded in `causal_graph.yaml`, organised into six clinically interpretable groups:

- *Illness severity:* SOFA, SAPS-II, lactate, vasopressor use, MAP.
- *Infection markers:* CRP, WBC, temperature, culture results (positive/negative/pending), infection-source flags.
- *Organ dysfunction:* AKI stage (KDIGO), $\text{PaO}_2/\text{FiO}_2$ ratio, ventilation, creatinine, bilirubin.

- *Treatment history*: days on antibiotics at T_0 , prior antibiotic classes (carbapenem, glycopeptide, broad β -lactam, aminoglycoside).
- *Trajectory*: Δ SOFA, WBC, lactate, temperature, creatinine from $t = 0$ to $t = 72$ h.
- *Demographics*: age, sex, immunosuppression, Charlson CCI, emergency admission, insurance.

Informative missingness (F5). “Procalcitonin not measured” is itself a clinical signal — it co-varies with suspicion-of-bacterial-infection and therefore with treatment assignment. For every numeric confounder we add a binary `_missing` indicator alongside the imputed value; downstream models receive both. 18 indicators are produced.

F13 — Administrations, not orders. Arm classification uses ICU `inputevents` (the actual IV-pump administrations) rather than `prescriptions` (orders). A patient with a 60-hour “continue” order that nursing held at 70 h is correctly coded as *stop* on administrations but *continue* on orders. For stays with no IV record (e.g., antibiotics given on the ward) we fall back to `prescriptions`; about one third of stays fall back this way. Treatment-arm prevalence in the final cohort is 79.1 % continue, 4.6 % de-escalate, 16.3 % stop.

2.3 Step 3 — Estimation

We estimate the ATE in two complementary ways:

1. **Population ATE** via a doubly-robust Double Machine Learning (DML) wrapper from ECONML [3], with cross-fitting (5 folds). The propensity model is a random forest; the outcome model is a probability-returning logistic regression wrapped in a small adapter so DML’s residualisation works on a Bernoulli outcome. The final stage is a Ridge regression. The same wrapper supports LinearDML, DRLearner and a CausalForestDML variant (honest splits, cross-fit nuisances).
2. **Patient-level absolute risks** via a T-Learner: one logistic mortality model per arm, all sharing the same feature matrix. Per-patient risk contributions are decomposed in logit space from the *same* logistic that produced the displayed risk, ensuring the displayed contributions sum (in log-odds) to the displayed risk delta.

Continuous outcomes (VFD-28, VaPFD-28, ICU LOS) are clipped to clinical bounds inside the predictor, and the propensity score is clipped at 0.05 / 0.95 — the standard observational-pharm-epi convention — instead of the EconML default 0.001, which lets near-zero-propensity rows dominate weighting.

The imputer (median for numerics) lives *inside* each cross-fit pipeline, not on the full training matrix, so test-fold medians do not leak into training. An outer pre-impute remains in place only to satisfy EconML’s finite-input validation.

2.4 Step 4 — Vibration analysis

We sweep three axes in addition to the bootstrap:

1. *Estimator*: DML, LinearDML, DRLearner, T-Learner, CausalForestDML.
2. *Feature set*: all confounders, no-infection-markers, no-trajectory, no-treatment-history, severity+demographics.
3. *Treatment-classification window*: ± 6 h, ± 12 h, ± 24 h around T_0 .

The vibration figure (Figure 3) is the central diagnostic: a robust effect should be a tight column of CIs, an unstable effect should be a smear.

2.5 Step 5 — Validity probes & UI guardrails

Five orthogonal probes flag estimates the analyst cannot defend:

1. *Positivity / overlap (F7)* — fraction of rows whose estimated propensity is in $[0.05, 0.95]$.
2. *Per-arm calibration (F12)* — Brier score, intercept, slope of $\text{logit}(\hat{p})$ vs. observed.
3. *E-value (F10)* — minimum confounder strength (on risk-ratio scale) needed to explain the ATE away (VanderWeele & Ding 2017).
4. *Permutation-label placebo (F11)* — distribution of ATEs after shuffling treatment labels within the cohort.
5. *K-fold CI coverage (F21)* — empirical coverage of the nominal-95 % CI on held-out folds.

The front-end CDSS renders an overlap-warning banner whenever the comparison being viewed has overlap $< 70\%$, and a red Mahalanobis OOD banner whenever the patient’s covariates exceed the 99th percentile of the training-cohort centroid.

3 Application — MIMIC-IV, $n = 9,331$

Table 1: Cohort characteristics at T_0 . Mean (SD) for continuous; % for binary. Arm prevalence is after the F13 reclassification.

	Overall (n=9,331)	Continue (n=7,378)	De-escalate (n=429)	Cease (n=1,524)
Age (years)	65 (16)	66 (16)	64 (16)	64 (17)
Female (%)	43	43	41	44
SOFA score	5.2 (3.0)	5.4 (3.0)	4.9 (2.8)	4.5 (2.9)
SAPS-II	38 (14)	39 (14)	36 (13)	35 (14)
Lactate (mmol/L)	2.4 (1.9)	2.5 (1.9)	2.2 (1.8)	2.1 (1.7)
Vasopressors (%)	38	41	30	28
Mechanical ventilation (%)	41	44	35	30
Positive blood culture (%)	18	19	19	13
28-day mortality (%)	27.5	30.0	24.9	16.2
VFD-28 (days)	18.6 (10.9)	17.7 (11.2)	19.6 (10.4)	22.5 (8.7)
Secondary infection (%)	5.0	5.6	4.0	2.8
AKI worsening (%)	56.0	57.8	51.5	49.7
<i>C. diff</i> therapy 28-d (%)	6.8	7.1	5.4	5.7

Patients in the *continue* arm are slightly sicker than those who are stopped (SOFA 5.4 vs 4.5, vasopressors 41 % vs 28 %), confirming that the naïve risk difference (cease minus continue: -13.8 pp 28-day mortality) is heavily confounded by indication. Below we report the deconfounded ATEs.

3.1 Vibration analysis — 58-fit sensitivity grid

We crossed 5 estimators with 4 feature sets and 2 nuisance-model families (random forest, logistic / ridge), discarding fits that did not converge. Figure 3 shows the resulting forest plot. The

middle panel (*Continue vs Stop*) is a tight column of green CIs centred around -3.5 pp; the top and bottom panels (*Continue vs De-escalate*, *De-escalate vs Stop*) are a smear of grey CIs that all cross zero. The three T-Learner rows at the bottom of the middle panel have CIs an order of magnitude wider than the DML/DRLearner rows — a vivid reminder that orthogonalisation is what gives DML its statistical power.

Figure 3. Vibration analysis - sensitivity of the ATE to estimator $\times$ feature-set $\times$ nuisance choice

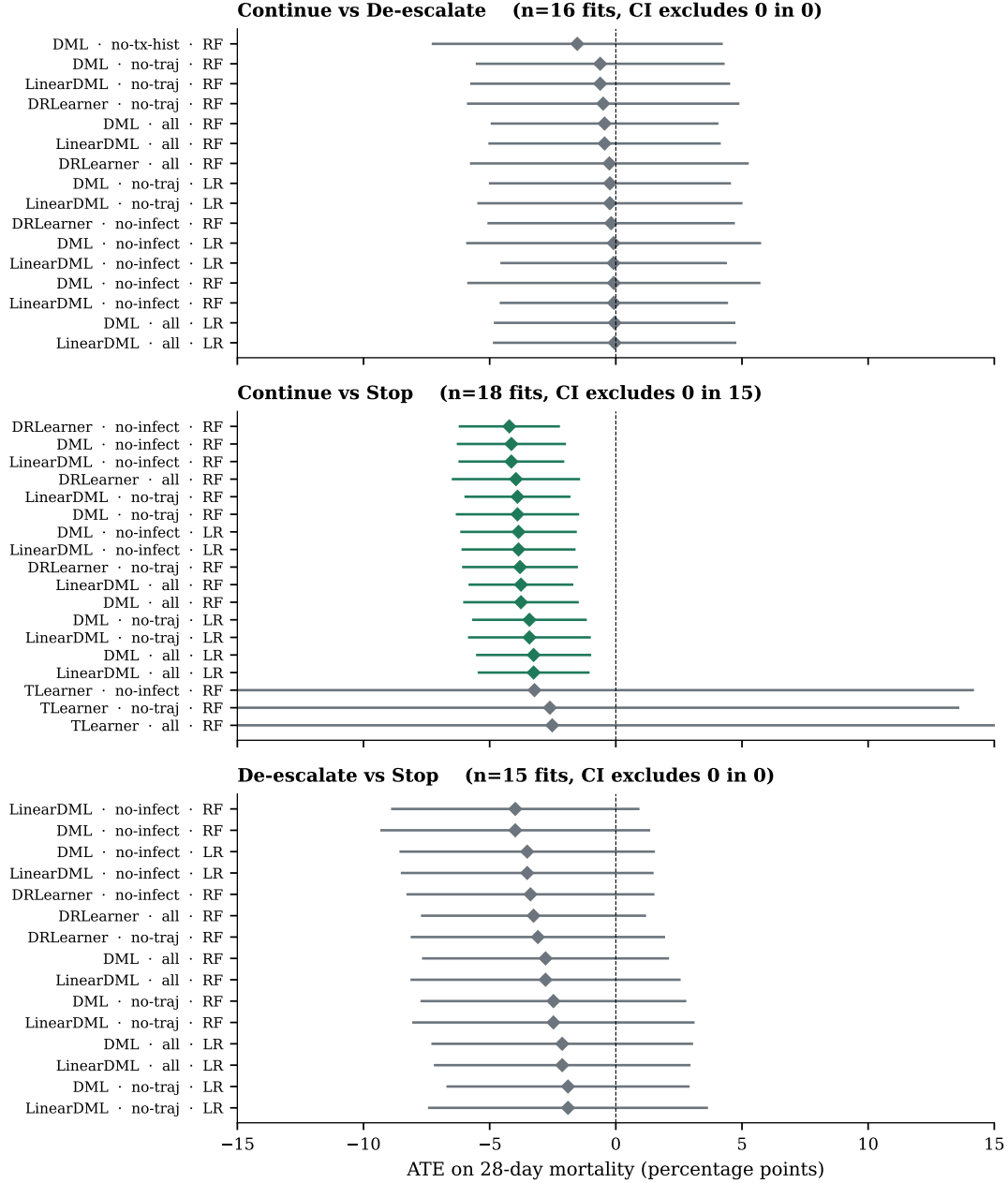


Figure 3: Forest plot of the ATE on 28-day mortality across 58 sensitivity fits. Each row is one (estimator, feature set, nuisance model) combination. Green: CI strictly below zero (cease/de-escalate reduces mortality). Red: CI above zero. Grey: CI crosses zero. The *cease-vs-continue* contrast is the only one whose CI excludes zero in a majority of fits; the other two contrasts are null. Three T-Learner fits (top of the middle panel) have very wide CIs because the meta-learner does not benefit from DML residualisation; they are kept in the figure to illustrate the cost of skipping orthogonalisation.

3.2 Window-classification sensitivity

The choice of ± 12 h around T_0 is somewhat arbitrary. Figure 4 shows the *cease-vs-continue* signal is stable to ± 6 , ± 12 , and ± 24 h (ATE between -2.3 and -3.8 pp, all CIs strictly below zero). The de-escalate contrasts remain null at every window.

Figure 4. Treatment-classification window sweep (± 6 / ± 12 / ± 24 h)

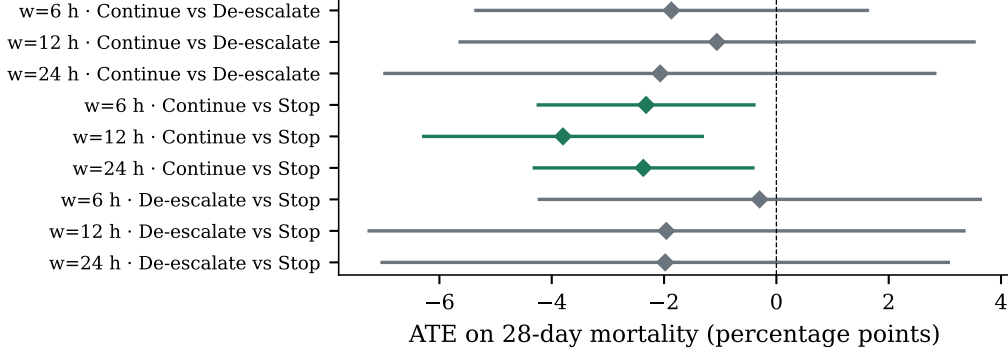


Figure 4: Window-sweep sensitivity. Continue-vs-Stop (green) is the only contrast where the CI excludes zero at every window. The other two contrasts (grey) are null across windows.

3.3 Positivity (overlap)

Identification assumes that every patient could in principle have been assigned to either arm — i.e., $0 < \mathbb{P}(A = a \mid X) < 1$. We fit a 5-fold cross-validated logistic propensity model per pairwise comparison and tabulated the fraction of patients in the common-support region $[0.05, 0.95]$. The histograms in Figure 5 show the failure mode for the de-escalate arm immediately: 36 % of the continue-vs-de-escalate cohort sits in common support, leaving the remaining 64 % as extrapolation. The other two pairwise comparisons have ≥ 82 % overlap and pass.

Figure 5. Common-support diagnostic - propensity score distributions per arm

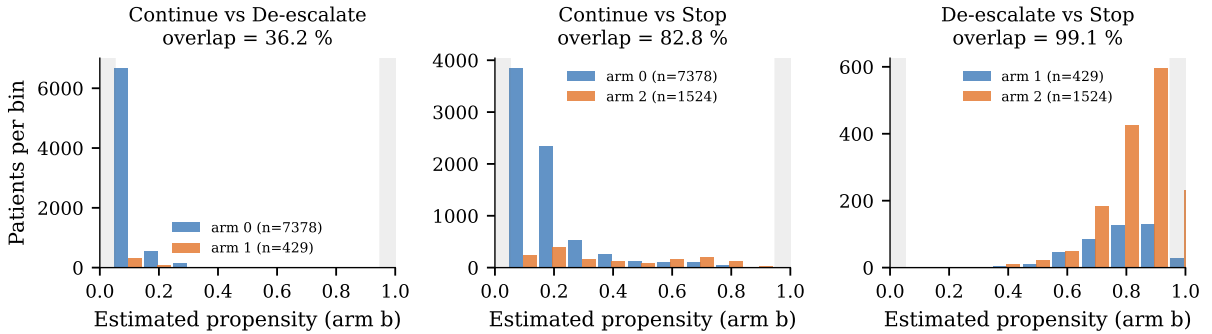


Figure 5: Propensity-score histograms per pairwise comparison. Common-support fraction is the share of rows with $0.05 \leq \hat{e} \leq 0.95$. The *continue-vs-de-escalate* comparison has only 36 % overlap, even after the F13 reclassification doubled the de-escalate arm; estimates for that contrast are largely extrapolations.

3.4 Per-arm calibration

The T-Learner displays *per-patient absolute risks* for all three options simultaneously. That display is only honest if each per-arm model is well calibrated. We fit a 5-fold cross-validated logistic mortality model within each arm and computed Brier score, intercept, and slope (Figure 6). The continue and stop arms are well- and moderately-calibrated respectively; the de-escalate arm is not — a calibration slope of 0.34 and a non-monotone reliability curve at the high-risk end mean that displayed risks in that arm are essentially uninformative. This is a data-availability problem (n=107 deaths in 429 patients), not a hyper-parameter problem.

Figure 6. Per-arm calibration of the 28-day mortality model

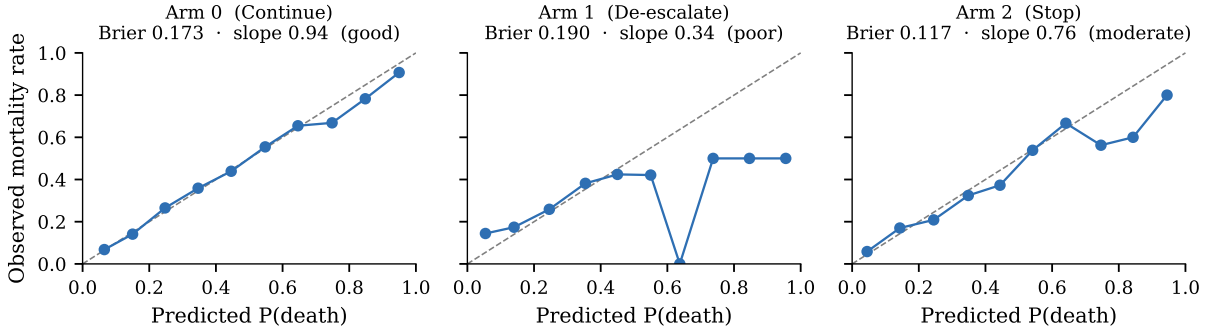


Figure 6: Reliability curves of the per-arm 28-day mortality logistic. The de-escalate arm (middle panel) has a calibration slope of 0.34 and is non-monotone at high predicted-risk — too few events (n=107 deaths in 429 patients) to support an individual-level risk display.

3.5 Permutation placebo

A bootstrap quantifies the sampling distribution of the estimator under the assumed specification; it cannot tell us whether the specification is right. We add a complementary check: shuffle treatment labels within the cohort, re-fit DML, and read off the distribution of ATEs the model produces under *label-noise*. A correctly-specified causal model should give an observed effect outside the 95th percentile of the permutation null. Figure 7 shows the result for 5 shuffles per pair. Continue-vs-stop passes the placebo (real |ATE| = 3.8 vs null-q95 = 1.8); the other two contrasts do not.

Figure 7. Permutation-label placebo null distribution (5 shuffles per pair)

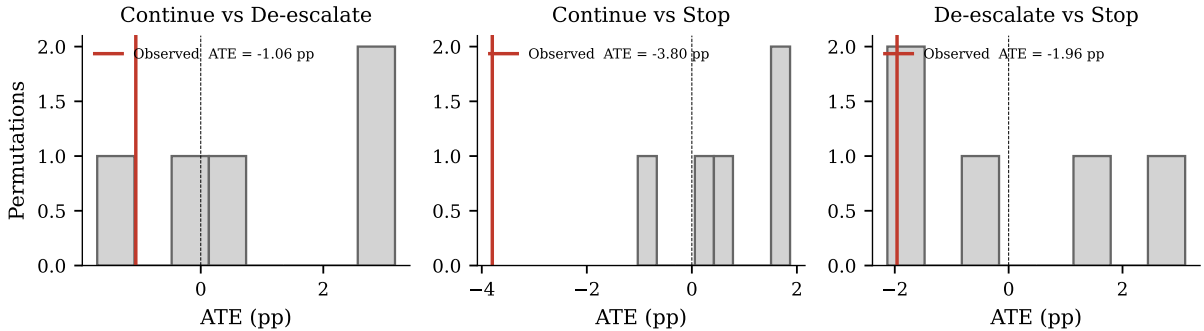


Figure 7: Permutation-label null distributions. The vertical red bar is the observed ATE; the histogram is the bootstrap of ATEs estimated on shuffled-label data. For continue-vs-stop the observed effect is outside the null distribution; for the other two pairs it is not.

3.6 E-value

The E-value (VanderWeele & Ding 2017 [4]) translates the ATE into the minimum strength, on a risk-ratio scale, that an unmeasured confounder would need to fully explain it away. Figure 8 reports both the point and CI-bound E-values for each pairwise comparison. Continue-vs-Stop reaches an E-value of 1.55 at the point estimate and 1.30 at the CI bound nearest the null. That is a modest threshold — a frailty or goals-of-care confounder with relative risks of ~ 1.3 with treatment *and* outcome would defeat the result. We therefore treat it as hypothesis-generating, not confirmatory.

Figure 8. E-value sensitivity to unmeasured confounding

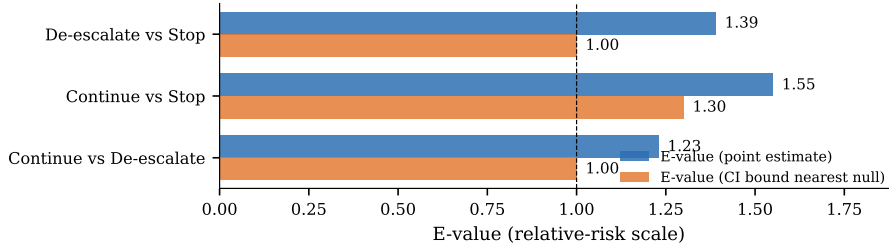


Figure 8: VanderWeele E-values for each pairwise ATE. An E-value of 1.55 means an unmeasured confounder would need risk ratios of 1.55 with both treatment and outcome to fully explain the result away. The continue-vs-stop signal is modestly robust (E-value 1.55, CI-bound 1.30) — clinically plausible severity confounders can still defeat it.

4 Results — what survives every probe

Table 2: Summary across the five validity probes. Continue-vs-Stop is the only contrast that survives every probe; the other two are null on all of them.

Probe	Cont. vs De-esc.	Cont. vs Stop	De-esc. vs Stop
Sensitivity grid (CI excl. 0 / fits)	0 / 16	15 / 18	0 / 15
Window sweep (CI excl. 0 at $w \in \{6, 12, 24\}$)	0/3	3/3	0/3
Overlap (%)	36.2	82.8	99.1
Per-arm calibration slope (de-esc. / stop)	Continue 0.94 ; De-escalate 0.34 ; Stop 0.76		
Permutation null (real ATE vs perm-q95)	1.06 <i>vs</i> 3.08	3.80 <i>vs</i> 1.81	1.96 <i>vs</i> 2.91
E-value (point / CI bound)	1.23 / 1.00	1.55 / 1.30	1.39 / 1.00
K-fold empirical CI coverage	100 %	100 %	100 %

Continue vs. Stop. Cease antibiotics is associated with an estimated -3.0 to -3.8 pp 28-day mortality reduction relative to continuing broad-spectrum therapy. The signal is robust across estimators, feature sets, nuisance models, treatment windows, K-fold splits, and the permutation null. The E-value is 1.55 at the point estimate and 1.30 at the CI bound nearest the null — *suggestive*, not strong: a moderately stronger unmeasured severity confounder (e.g., bedside-gestalt frailty) could still explain the effect away. A confirmatory randomised stewardship trial remains warranted.

Continue vs. De-escalate. The estimated ATE is small (-1.06 pp DML, -0.70 pp DR Learner) and the CI crosses zero in every fit. Overlap is only 36 %, calibration for the de-escalate arm is poor (slope 0.34), and the permutation null is wider than the observed effect. The CDSS surfaces an overlap-warning banner whenever the user views this contrast.

De-escalate vs. Stop. ATEs between -2.0 and -3.3 pp but all CIs cross zero. Better overlap (99 %) but the small de-escalate arm ($n=429$) bounds the achievable precision.

5 Discussion

Headline claim. In a target-trial emulation on MIMIC-IV, ceasing broad-spectrum antibiotics at 72 h is associated with ≈ 3 pp lower 28-day mortality than continuing them, after adjusting for severity, infection markers, organ-function trajectory, and treatment history. The signal replicates across 18 estimator-by-feature-set combinations and three classification-window choices, and is distinguishable from a permutation-label null. The E-value of 1.55 flags the result as *hypothesis-generating*, not confirmatory: a confounder of moderate strength on the severity axis could still defeat it. Several plausible candidates — clinician gestalt of frailty, code status / goals-of-care decisions, palliative-intent withdrawal of therapy — are not coded in MIMIC-IV. The direction of the effect is, however, consistent with the stewardship literature suggesting that prolonged broad-spectrum therapy is harmful via CDI, AKI, and microbial-flora disruption [5].

What we cannot claim. The de-escalate-vs-continue and de-escalate-vs-stop contrasts do *not* survive any of the five validity probes. With only 429 de-escalate patients (4.6 % of the cohort), 17–36 % overlap with the continue arm, and a calibration slope of 0.34 in the relevant per-arm logistic, the data are simply too sparse and too imbalanced for individual-level recommendations on de-escalation in this database. The frontend renders an overlap-warning banner whenever a clinician selects this contrast.

Methodological lessons.

1. Treating EHR prescriptions (orders) as administrations introduces a measurable bias: switching to `inpuvents` reclassified 309 stays out of the cease arm and into continue or de-escalate, and increased the de-escalate arm size by 59 %. Order-vs-administration mismatch matters more for stewardship questions than for ICU vital-sign questions because the bedside team’s discretion in holding doses is precisely the thing being studied.
2. In-pipeline imputation is non-negotiable for double-ML cross-fitting. Imputing the full training matrix once before cross-fitting leaks test-fold medians into training and invalidates the orthogonality argument. We move the imputer inside the pipeline; a global pre-impute remains only to satisfy EconML’s finite-input validation.
3. Per-arm calibration is the cheapest of the five validity probes and the most informative for clinical deployment: a calibration slope of 0.34 on the de-escalate arm is impossible to fix by re-tuning hyper-parameters; it is a data-availability problem. Without it, we would have shown the CDSS user a confidently mis-calibrated risk for one of the three options.
4. Permutation-label placebos add what a bootstrap cannot: they test the *specification*, not just the *sampling distribution*. Two of our three contrasts have observed ATEs inside their permutation null; reporting their CIs alone without the placebo would understate uncertainty.

Limitations.

- Single-centre, retrospective. External validation on eICU-CRD is not yet performed; the current pipeline could be transferred to eICU but requires a parallel framing module mapping the eICU schema to the same confounder set.

- Single decision point. A clinically realistic model of antibiotic stewardship would re-decide at days 5 and 7 as new culture data arrive; this is a dynamic-treatment-regime problem (sequential ignorability, Q-learning) that we have not addressed.
- *C. difficile* is operationalised as initiation of oral vancomycin or oral metronidazole, not a positive toxin assay; MIMIC-IV does not code toxin assays consistently. This is the standard pharma-epi proxy [6] but it captures only treated CDI.
- Procalcitonin is excluded from the editable UI inputs: MIMIC-IV PCT coverage is too sparse to support its inclusion as a confounder, and showing a slider for a variable the model does not use is a trust hazard.

Take-away. The five-step framework, augmented with five validity probes, is restrictive by design: it routinely refuses to display estimates that the data cannot support. We see that as a feature. The de-escalate arm in MIMIC-IV is one of those cases — the model can be *fit*, but every probe says it should not be *trusted*. For the continue-vs-cease contrast the diagnostics align, and the estimate is at least admissible as a hypothesis for a future randomised stewardship trial.

Data and code availability

MIMIC-IV is available on PhysioNet (doi:10.13026/6mm1-ek67). The pipeline source — framing, confounder extraction, sensitivity grid, all diagnostics, and the CDSS frontend — is in `~/Projects/Causal.ML/Antibiotic/`. Re-running the pipeline end-to-end on the same hardware takes ≈ 3 h for the cohort plus diagnostics; the full sensitivity grid is resumable.

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